

Revenue Leakage Intelligence System

A Multi-Source Analytical Framework for Quantifying and Prioritising Revenue Loss

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The Problem Nobody Is Measuring

Every business loses revenue silently — not through bad sales, but through operational gaps that go undetected.

Discount Abuse

Discounts applied beyond the breakeven threshold

Pricing Gaps

Products sold below their established fair-market price

Billing Shortfalls

Payments collected that fall short of expected order value

Returns Fraud

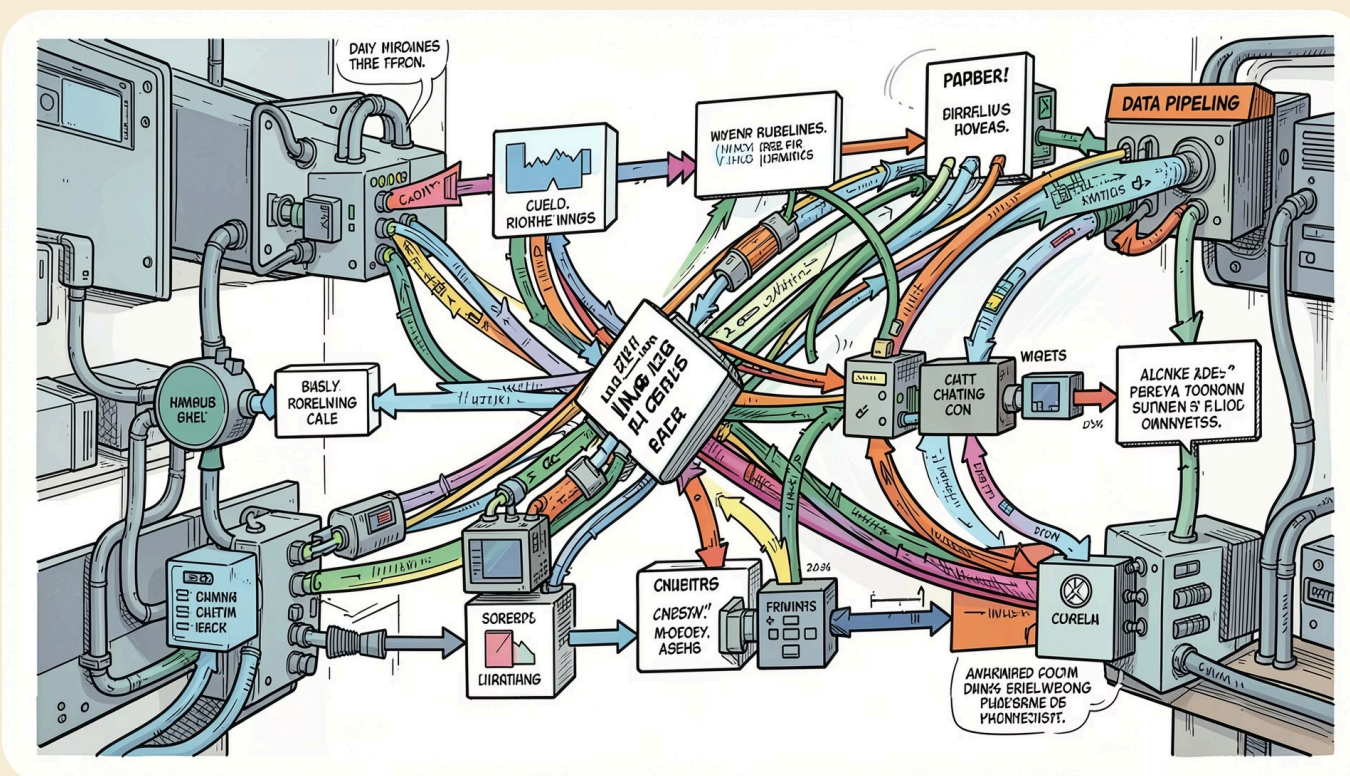
Goods returned at rates that indicate abuse or fraud

These four channels collectively represent a quantifiable, recoverable revenue gap — if you know where to look.

"You can't fix what you can't measure. This project builds the measurement framework."



Project Overview



The **Revenue Leakage Intelligence System** is an end-to-end analytical pipeline that:

Tech Stack: Python · Pandas · NumPy · SQLite · Matplotlib · Seaborn · Power BI

Pipeline Structure: Notebook 01 (Superstore) » Notebook 02 (Olist) » Notebook 03 (UCI) » Notebook 04 (Master KPI) » Power BI Dashboard

- ➔ Ingests **3 real-world datasets** from distinct business contexts
- ➔ Applies **domain-specific leakage detection logic** to each
- ➔ Consolidates all findings into a **unified KPI summary**
- ➔ Delivers an **executive Power BI dashboard** with 3 analytical views

The Four Leakage Types Defined

Leakage Type	Definition	Dataset Used
Discount Leakage	Revenue lost when discounts exceed 20% AND cause a net loss	Superstore (US Retail)
Pricing Leakage	Revenue foregone when products are sold below their 90th percentile benchmark price	Superstore (US Retail)
Billing Leakage	Payment shortfall where collected amount is less than expected (product + freight)	Olist (Brazil E-Commerce)
Returns Leakage	Revenue reversed through product returns — especially from high-rate customers	UCI Online Retail II (UK)

Dataset Architecture

Three Real-World Datasets. Three Business Contexts. One Unified View.

Dataset 1 — Sample Superstore (Kaggle)

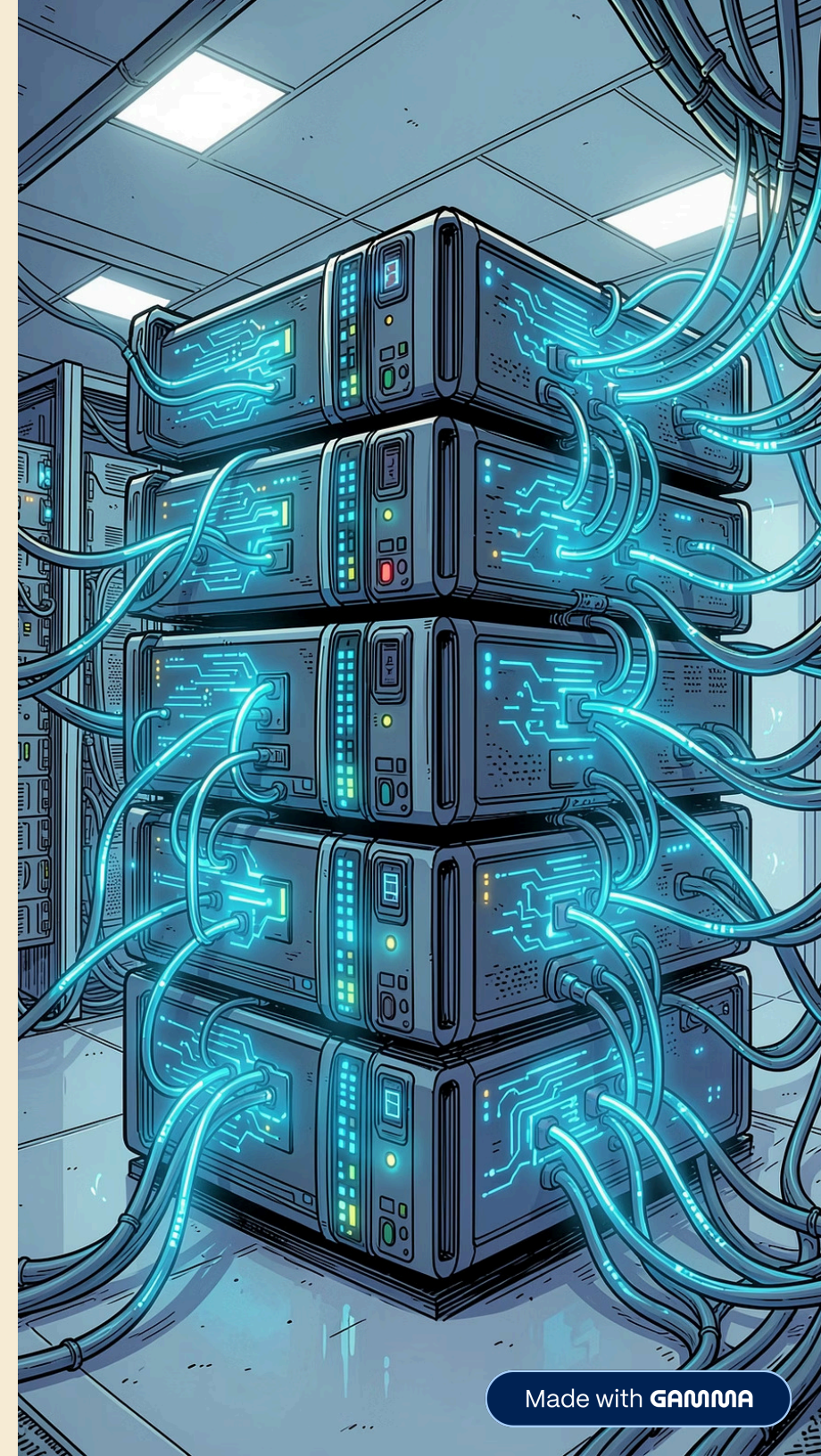
- ~9,994 transactions | US Retail | 2014–2017
- Covers: Furniture, Office Supplies, Technology
- Used for: **Discount Abuse** + **Pricing Leakage**

Dataset 2 — Olist Brazilian E-Commerce (Kaggle)

- 99,441 orders + 112,650 items + 103,886 payment rows
- Multi-table relational structure (Orders + Items + Payments)
- Used for: **Billing Reconciliation Gaps**

Dataset 3 — UCI Online Retail II

- ~1,067,371 transaction rows | UK Wholesale | 2009–2011
- Negative-quantity invoices = confirmed return signal
- Used for: **Returns Leakage** + **Customer Risk Segmentation**



Methodology — Notebooks 01 & 02

NOTEBOOK 01 — SUPERSTORE

Discount & Pricing Leakage

Discount Leakage Detection:

- Flag condition: Discount % > 20 AND Profit < 0
- Leakage amount: abs(Profit) for all flagged transactions

Pricing Leakage Detection:

- Benchmark price = 90th percentile Unit Price per Product ID (robust, outlier-resistant)
- Leakage amount = (Benchmark Price – Actual Unit Price) × Quantity, clipped at zero

Key Engineering Steps:

- Unit Price derived as Sales / Quantity
- Profit Margin % = (Profit / Sales) × 100
- Loss Flag and High Discount Flag added as binary indicators

NOTEBOOK 02 — OLIST

Billing Reconciliation

Three-Table Join Architecture:

```
orders LEFT JOIN items_aggregated LEFT JOIN  
payments_aggregated ON order_id
```

Feature Engineering:

- `expected_payment` = SUM(price) + SUM(freight_value) per order
- `payment_gap` = expected_payment – total_payment_value

Leakage Flagging:

- Threshold: payment_gap > 1 (eliminates floating-point processor noise)



Why This Matters: At scale, even small per-order shortfalls compound into significant unreconciled revenue.

Methodology — Notebooks 03 & 04

NOTEBOOK 03 — UCI

Returns & Customer Risk

Return Signal Identification:

- Primary: Quantity < 0 in transaction log
- Verified via: Invoice prefix = "C" (cancellation invoice) — confirms these are genuine returns, not data entry errors

Return Amount Calculation:

- Return Amount = abs(Quantity) × Price

Customer Risk Segmentation Model:

Return Rate %	Risk Category
≤ 5%	Low Risk
5% – 15%	Medium Risk
> 15%	High Risk

NOTEBOOK 04 — MASTER KPI

The Consolidation Layer — Turning Analysis into Strategy

All three upstream tables loaded from SQLite into a single session.

Aggregation Logic:

- discount_leakage = SUM(Discount Abuse Amount) where > 0
- pricing_leakage = SUM(Pricing Leakage Amount) where > 0
- billing_leakage = SUM(Billing Leakage Amount) where > 1
- returns_leakage = SUM(Return Amount) where > 0

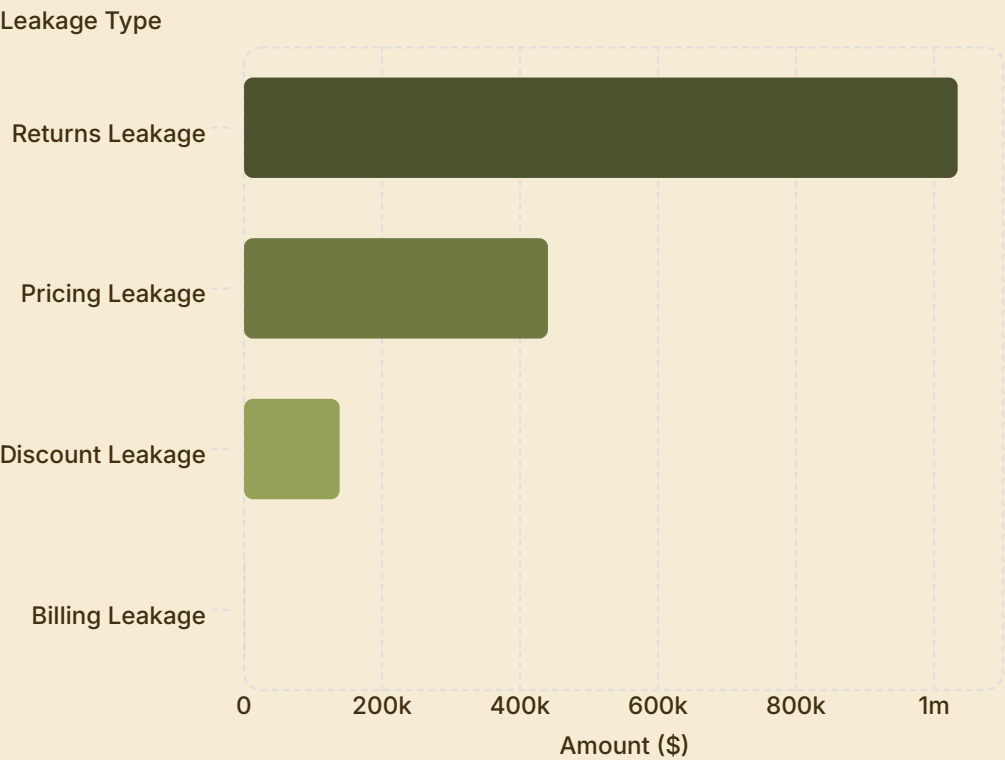
Severity Classification:

- Contribution < 10% » **Low**
- Contribution 10–30% » **Medium**
- Contribution ≥ 30% » **Critical**

Output: Master KPI table stored to SQLite + 4 CSVs exported for Power BI

Key Findings — The Numbers

Total Revenue Leakage Identified: \$1,613,323



Leakage Type	Amount	Severity
Returns Leakage	\$1,034,086	Critical
Pricing Leakage	\$440,381	Medium
Discount Leakage	\$138,515	Low
Billing Leakage	\$341	Low



The headline insight: Returns leakage alone accounts for nearly **two-thirds** of all identified revenue loss — yet it is the leakage type most businesses have the **least visibility into**.



Power BI Dashboard — 3 Analytical Views

The executive-facing dashboard transforms raw leakage data into actionable intelligence across three distinct analytical perspectives.

1

View 1 — Executive Summary

- Big KPI cards showing total leakage per type
- Severity color coding (Critical / Medium / Low)
- Percentage contribution breakdown

2

View 2 — Operational Drill-Down

- Discount leakage by Category and Region
- Pricing leakage by Product Sub-Category
- Customer risk distribution chart

3

View 3 — Transaction-Level Detail

- Filterable tables for each leakage type
- Supports root-cause investigation
- Exportable for operational follow-up

Business Impact & Conclusion

Why This System Matters



Detects Hidden Losses

Identifies revenue leakage invisible to standard financial reporting



Prioritizes by Severity

Focuses remediation efforts on the highest-impact leakage channels first



Prevents Future Loss

Customer risk scoring helps prevent repeat return abuse before it compounds



Scalable Framework

Multi-dataset architecture allows seamless extension to new data sources and leakage types

Revenue leakage is not a finance problem — it is a data visibility problem. This system closes the gap.

